AnkGuru (अंकगुरू): Field Study & System Design Report

An offline app that helps rural tribal children learn numbers and mental math

ET 617: Educational Application Development | IIT Bombay

Team	Group 5
Members	Lakshaditya Singh (23B2230) Aagam S. Kasaliwal (23B2215) Aditya Anand Gupta (23B2144)
Client	N+1 NGO
Primary contact	Mr. Amod Joshi, Founder & Coordinator
Where it will be used	Ashram and Zilla Parishad schools, Palghar district
Motto	“One Story. One Idea. One Action”

Executive Summary

In Palghar’s tribal schools, many children in Grades 3 to 6 follow spoken Marathi numbers easily but struggle to connect them to written Devanagari digits. One teacher looks after a large multi-grade class, so feedback is slow, and there is no internet to fall back on. AnkGuru is a fully offline Android app made for cheap, shared phones. A child hears a number in Marathi, then picks it, draws it on two chalkboard slates (tens and units) that a small 156k parameter CNN reads, or says it aloud and checks it against the app’s audio. Mr. Amod Joshi tested it live with his learners at his Palghar centre and shared the results with us. The children enjoyed it, especially scribble mode, and feedback arrives instantly with no waiting for the teacher. Splitting tens and units across two slates also targets the common 28 versus 82 reversal. The latest release, v3.2 Responsive, is 55.46 MB and costs nothing to run.

Understanding of Requirement & Context

The problem. N+1 NGO works with first generation learners, mostly from the Katkari and Warli communities, in Zilla Parishad and Ashram schools. Their brief to us was simple. Children don’t get personal, fear-free numeracy practice beyond the blackboard, so they start avoiding math and the learning loss keeps growing.

What the setting looks like

- **Learners:** Most are the first in their family to attend formal school, from Katkari and Warli communities. They speak Marathi fluently, but many can’t link a spoken quantity to its Devanagari digits.
- **Classrooms:** One or two teachers run multi-grade rooms of 35 to 45 students (Grades 3 to 6), so individual attention and quick correction aren’t realistic.
- **Infrastructure:** No reliable mobile data or Wi-Fi, frequent power cuts, and only shared low-end Android phones (Android 8 to 14, 2 to 3 GB RAM) handed out by facilitators.

What the system has to do

1. **Ground numeracy in spoken Marathi.** Prompts are audio first (mr-IN, falling back to hi-IN), so children never have to read instructions.
2. **Work 100% offline.** Number recognition and speech both run on the phone, inside one small APK.
3. **Build number writing through touch.** Separate Tens and Units slates, with the child’s drawing read by a CNN.
4. **Keep feedback fear free.** Instant, encouraging audio and visuals, with retries and no penalty for mistakes.

Field Study Methodology & Data Collection

Field Study Approach

We did not travel to Palghar ourselves. Mr. Amod Joshi runs the on-ground work with the learners, so we did the field study remotely through him, using four methods:

1. **Client interviews and calls.** Mr. Joshi described the classrooms, the shared phones, how multi-grade sessions run, and how children behave when they practise numbers.
2. **Client-led live testing.** We sent builds (v3.1, then v3.2) and Mr. Joshi tried them with his students at his centre. He told us where children hesitated, how they wrote their digits, and what they enjoyed.
3. **Client feedback video.** Mr. Joshi gave his feedback in his own words, covering what worked, what needed fixing, and how he sees the app fitting into his sessions.
4. **Our own device testing** on low-spec Android phones, comparing stroke heuristics against the CNN, and on-device speech models against local audio loops.

Data Collected

The classroom side below comes from Mr. Joshi, who tested the app live with his students and told us what he observed. The technical numbers are measured from our own builds and device tests.

Parameter	Traditional classroom	AnkGuru trial	What Mr. Joshi reported
Student attention and turns	One child at a time with the teacher, the rest wait	One to one, self-paced on a shared handset	Continuous individual practice without queuing
Practice volume	Only a few questions per child per session	Continuous sets of questions on the phone	Many more active problem solving turns
Feedback delay	Delayed, marked in batch at the end	Instant chime and visual highlight	Mistakes get caught before they stick
Digit reversal (e.g. 28 written as 82)	Common, children follow spoken word order	Tens (दशक) and Units (एकक) slates are separate	Place value is enforced before digit recognition
Engagement	Attention drifts during blackboard copying	Children stay with it, scribble mode is the favourite	Children treat the slate as play

Measured technical data	Result	Why it matters
APK size	About 198 MB first build, 55.46 MB now	Installs on shared phones over Bluetooth or Share
Digit recognition	156k parameter CNN, under 50 ms on device	No lag on entry-level phones
Whisper.rn and IndicWav2Vec	8 to 14 seconds per inference, overheating	Led us to drop on-device speech recognition

Key Insights & Takeaways

- **Writing needs hands.** Passive tapping doesn't stick. Mr. Joshi reported that children treat the slate (पाटी व पेन्सिल) like a game, and that is what held their attention.
- **On-device speech recognition didn't work.** Whisper.rn and AI4Bharat IndicWav2Vec took 8 to 14 seconds per inference on budget Android phones, made them overheat, and pushed the APK past 198 MB. We dropped general speech recognition.
- **Children's digits are messy,** especially १, २, ४, ५, ८ and ९. Early handwriting trials on children's numerals failed, and template matching was too unreliable to use. What worked was 2D integral bounding box normalisation, a CNN retuned with synthetic data, and accepting the CNN's runner-up guess when it scores 20% or more.
- **Facilitators need zero setup.** Going by what Mr. Joshi told us about his facilitators, logins, passwords, syncing and settings would lead to abandoned sessions, so the app opens straight into practice.

User Persona & Scenario Mapping

Primary User Persona: Meera, Student

Representative persona, built from what Mr. Joshi told us about his learners. Age 9 to 10 | Grade 4 | Ashram residential school, Palghar | First generation learner (Warli community), native Marathi speaker, no smartphone or internet at home.

Role and goals	Student. Wants to feel confident with Devanagari numbers (1 to 100) and mental addition and subtraction, finish exercises without fear of scolding or embarrassment, and keep up with the class.
Pain points	Can't read text-heavy instructions. Mixes up number names (for example "सत्तेचाळीस", 47) when writing digits. Gets anxious solving math aloud in front of the whole class.
Technical literacy	Very low. Can't type or move around in settings, but understands large high-contrast cards, voice prompts and finger-drawing slates.
Environment	Shared budget Android phone (2 GB RAM, cracked screen), no mobile data, and a noisy classroom full of chatter and ceiling fans.

Secondary persona: a community learning facilitator (N+1 NGO). Basic smartphone user (WhatsApp and calls). Runs a daily practice session for a group of students across grades, and needs to spot who needs one-on-one help without marking papers. Can't fix software bugs, manage student accounts, or count on school internet.

User Journey: A Typical Practice Session

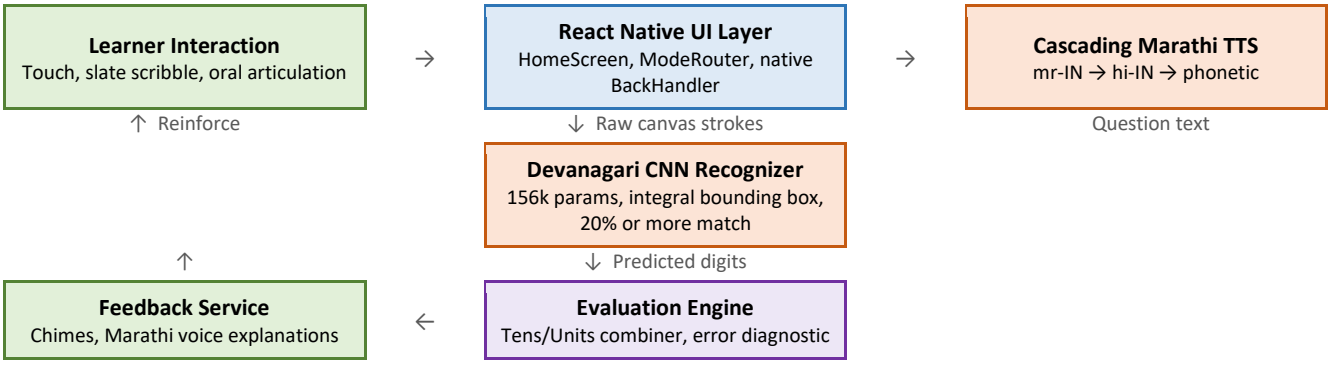
Step	What Meera does	What AnkGuru does	How she feels
1. Launch	Gets the shared phone and taps the orange AnkGuru icon.	Opens quickly, no login.	Curious, in control.
2. Choose practice	Picks Level 2 (11 to 50) and a 10 question set.	Highlights the level, updates the counter.	Sets her own target.
3. Listen	Opens Listen & Draw and hears "ऐका आणि लिहा: ३६".	Plays the number in Marathi through on-device TTS.	Quietly repeats "छत्तीस" to herself.
4. Draw	Writes ३ on the Tens slate and ६ on the Units slate.	Captures strokes, plays a chalk sound.	Hands-on, learns place value.
5. Check	Taps the green "तपासा" button.	Scales the strokes and the CNN reads both digits almost instantly (a runner-up at 20% or more is accepted).	Shaky writing isn't punished.
6. Feedback	Sees a green banner and hears a chime: "अचूक उत्तर: ३६, खूप छान!"	Plays praise in Marathi, then moves on to the next question.	Rewarded, less anxious.
7. Speak & Verify	Sees 14 + 6 = ?, says "वीस" aloud, taps "उत्तर ऐका".	Plays "वीस" in Marathi. The facilitator confirms.	Checks her own pronunciation.

Design Decisions & System Architecture

Design Rationale for Persona

Design choice	Constraint it addresses	How it is built
3-tier Marathi TTS	Meera can't read English or dense text, and a phone may not have a Marathi voice.	Checks Android's TTS service and falls back to mr-IN, then hi-IN, then phonetic.
Dual chalkboard slates	Children mix up tens and units in two digit numbers.	Two separate 220px tall HTML5 canvases (Tens: दशक, Units: एकक) inside a React Native WebView.
Tolerant CNN (runner-up at 20% or more)	Immature fine motor control gives ambiguous, curvy strokes.	156k parameter CNN with 2D integral bounding box normalisation.
Speak & Verify	Heavy speech models freeze and crash low-end phones.	The child speaks, the app plays the correct audio, and the teacher confirms.
Native BackHandler	Children press the hardware back button and exit by accident.	A BackHandler hook steps back through screens instead of closing the app.

System Architecture & Data Flow



Everything above sits inside one standalone APK. Zero cloud, zero network traffic.

Everything runs on the device inside one standalone APK, with no cloud calls. There are three layers. **Presentation:** a multi-mode React Native dashboard, with hardware BackHandler managing navigation state. **Handwriting engine:** two HTML5 slates in a WebView that normalise the 2D bounding boxes into 32x32 matrices for 156k CNN inference (under 50 ms). **Speech pipeline:** a declared Android TTS service that cascades from mr-IN to hi-IN phonetic mapping, with a 500 ms settling delay.

Core Features & Edge Case Handling

- **Question generator:** Builds arithmetic questions across 4 levels (1 to 10, 11 to 20, 21 to 50, 51 to 100) and tracks past seeds so the same question never shows up twice in a row.
- **Audio settling delay:** A 500 ms pause between a screen change and speech fixes the clipped audio that showed up in testing when Android audio focus was interrupted before it was ready.
- **Answer leak fix:** Feedback state and canvases now reset in the same step as the next question, which removed a brief flash of the previous answer.
- **Responsive slate, zero screen drift:** We swapped the ScrollView for a locked flex:1 container (touch-action: none) with dynamic viewport height clamping, so the screen no longer scrolls while a child is drawing on compact phones like Realme and Redmi.

Field Impact, Client Validation & Next Steps

Client Validation & Feedback Summary

Mr. Amod Joshi tested AnkGuru v2, v3.1 and v3.2 live with his students at his tutoring centre in Palghar and shared the results with us. The children really enjoyed it, especially scribble mode. This is what he told us in the client video:

- **Scribble mode works.** "Students love the scribble mode because it's an activity-based item." He added that math is something you actively do, not just something you learn in a classroom.
- **Quick fixes when accuracy dipped.** He thanked the team for retuning the model with synthetic data after early handwriting trials failed on children's numerals.
- **Speak & Verify is a sensible compromise.** Full on-device speech recognition wasn't reliable on low-cost phones in this timeframe, but the teacher-supervised oral loop keeps the class moving without crashes.

Impact Metrics & Value Proposition

This comes from Mr. Joshi's live testing and our own build measurements. We did not run a formal measurement study, and the pilot in the roadmap below is where proper numbers will be collected.

Dimension	Manual baseline	AnkGuru v3.2 (Responsive)	Value to the NGO
Practice turns	Limited turns per child on the blackboard	Continuous self-paced sets on a handset	More practice with the same teacher workload
Error detection	End of period, after manual marking	Instant on-device feedback	Wrong methods don't harden
Hardware	Paper worksheets or PCs	Budget Android phones	Runs on shared, donated phones
Running cost	Data plans and printing	₹0 recurring	Sustainable with no cloud bill
App size	About 198 MB first build	55.46 MB	Roughly 70% smaller, easy to share over Bluetooth

Implementation Roadmap (planned, not yet observed)

1. **Weeks 1 to 2:** Roll out v3.2 Responsive to 3 pilot Zilla Parishad and Ashram schools in Palghar, with a target of about 200 learners, to be confirmed with Mr. Joshi.
2. **Months 1 to 3:** Run Mr. Joshi's planned quasi-experimental study, comparing AnkGuru and control groups, and log real practice and accuracy data for learning gains and effect size.
3. **Month 4:** Add on-device SQLite learner profiles to track accuracy trends and repeated confusions, still with no cloud sync.
4. **Months 5 to 6:** Train a compact keyword spotter (under 10 MB) for Devanagari numerals (०–९) to score oral answers automatically on low-end phones.

Links to Artifacts

- **GitHub repository:** https://github.com/EAD-Labs/Group-5_2026.git
- **Project introduction webpage:** **AnkGuru (अंकगुरू) — Offline AI-Powered Foundational Numeracy for Rural Tribal Learners**
- **Jira board tasks:** **Group-5_2026/Jira-tasks.pdf at main · EAD-Labs/Group-5_2026**
- **APK download (v3.2 Responsive, 55.46 MB):** https://drive.google.com/file/d/1aKDtSQNlxlnIVq79fnIzRep6o6aBYD3f/view?usp=sharing
- **Client feedback video (Mr. Amod Joshi):** https://drive.google.com/file/d/1ar5OOENPHX6zPU_vvquvzu9AqVEOTYfv/view?usp=sharing